Two necessary conditions on minimal-time solutions of the firing squad problem

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Abstract

We prove two necessary conditions on the space-time diagram of every minimal-time solution of the firing squad synchronization problem, whatever its number of states. Every line agrees with the infinite half-line until the echo of its right end returns, and the rest of the line is a function of two anti-diagonals of the half-line. First, a pumping argument at the right end shows that the non-periodic part of the half-line reaches the line $i = t/3$ behind the front. Second, near the left border the reflected part of the line acts as a finite-state transducer on the half-line and cannot time the firing along long periodic stretches; hence the half-line is not periodic in any wedge at the left border, and it is never eventually regular. Both conditions can be checked on computed half-lines. With computer assistance, including LRAT-certified unsatisfiability proofs, we use them to show that every five-state minimal-time solution uses at least 15 distinct neighbourhoods on its half-line below the anti-diagonal $t + i = 78$, to re-prove with a certificate that four states do not suffice, and to confirm a conditional five-state result claimed by Balzer in 1967. A five-state rule that synchronizes all lines of length at most 14 bounds what refutations of this kind can achieve.

Keywords: firing squad synchronization problem, cellular automata, state complexity, minimal time, pumping lemma, proof certificates

1. Introduction

The firing squad synchronization problem, posed by Myhill in 1957 and popularised by Moore [1], asks for a finite automaton that, placed in every cell of a line of arbitrary length n and started with all cells quiescent except a *general* at one end, drives all cells into a firing state for the first time at the same instant. Goto [2] and Waksman [3] solved it in time $2n - 2$, which is optimal. This note proves two necessary conditions on every minimal-time solution, whatever its number of states. In the frame of the front the cells at a fixed depth behind the front become periodic in time, but along infinitely many depths the non-periodic part reaches the line $i = t/3$ (Theorem 3.4). The half-line is not periodic in time and space on any wedge at the left border (Theorem 4.2); hence it is never eventually regular, i.e. it never consists of finitely many periodic regions separated by signals of rational speeds (Corollary 4.3).

The least number of states of a minimal-time solution is a benchmark question [4, 5]: Waksman used 16 states, Balzer 8 [6], Gerken 7 [7] and Mazoyer 6 [8]. Balzer reported a computer search excluding four states [6]; Yunès [9] repeated it and observed that the lines of lengths $2, \ldots, 9$ suffice, and Sanders [10] found Balzer's backtracking incomplete and confirmed the result with a corrected search. Whether five states suffice has been open since 1987. As there are finitely many k-state rules, k-state minimal-time solutions do not exist if and only if for some N no k-state rule synchronizes all lines of lengths $2, \ldots, N$ in minimal time, so a negative answer may consist of finitely many necessary conditions. (Existence is different: the set of minimal-time solutions is not recursively enumerable [11].)

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Our conditions rest on an elementary decomposition (Section 2): until the echo of its right border returns, the line of length n behaves like the infinite half-line, so a solution consists of one half-line diagram C_∞ and, for each n, a triangle R_n that is a function of two anti-diagonals of C_∞ (Lemma 3.1). The right end of R_n must fire at time $2n-2$, so the inputs of the triangles of two lengths must differ on an explicit cone (Lemma 3.3); this gives Theorem 3.4. At the left end, R_n is computed by a band of anti-diagonals that acts as a finite-state transducer on C_∞ (Lemma 4.1, with the explicit bound $(k-1)^{2s}\ell$); this gives Theorem 4.2. The two conditions complement each other: the half-line of two five-state rules attains the bound of Theorem 3.4 with equality but violates Lemma 4.1 (Remark 3.5 and Corollary 5.3). Section 5 applies them, with computer assistance, to four and five states: a certificate for the four-state bound (Theorem 5.1), a five-state rule for all lengths up to 14 (Proposition 5.2), the bound $c_{78} \geq 15$ on the complexity of the half-line of every five-state solution (Proposition 5.4), and Balzer's conditional result (Proposition 5.5).

Related work. Indistinguishability arguments, in which two situations that a solution cannot tell apart force some cell to fire too early, exclude minimal-time solutions with one-way channels (Mazoyer [11, Prop. 2] compares the lines of lengths 2, 3 and 4) and, through a repetition of a pair of states forced by the pigeonhole principle, for variants with sub-generals and some geometric configurations [12, 13] (we follow the description in [13]). Our pumping lemma applies this mechanism to the original problem and to all lengths, where it constrains solutions instead of excluding them. The eventual periodicity behind the front used in Theorem 3.4 is classical and underlies the work of Mazoyer and Terrier on signals [14, 15]. Lower bounds are known for restricted classes: no three-state solution and no four-state symmetric minimal-time solution exist for rings [16], and Settle's thesis [17], as described in [5], excludes four- and five-state solutions under additional constraints. Solutions with at most five states exist for restricted families of lengths such as the powers of two, some of them in non-minimal time [18, 19]. Propositional encodings of cellular automata have been used for reachability questions [20]. We are not aware of statements equivalent to Theorems 3.4 and 4.2 or Corollary 4.3 in the literature.

2. Lines and the half-line

Let S be a finite set of *states* containing three pairwise distinct states, the quiescent state L, the general G and the firing state F; the *working* states are $W := S \setminus \{\mathsf{F}\}$, the *auxiliary* states are $W \setminus \{\mathsf{L}, \mathsf{G}\}$, and $* \notin S$ denotes the border. A *rule* is a map $\delta : (W \cup \{*\}) \times W \times (W \cup \{*\}) \to S$ with

$$\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \delta(\mathsf{L}, \mathsf{L}, *) = \mathsf{L}. \tag{1}$$

The *line of length* n is $C_n : \mathbb{N} \times \{1, \dots, n\} \to S$ with $C_n(0,1) = \mathsf{G}$, $C_n(0,i) = \mathsf{L}$ for $i \geq 2$ and $C_n(t+1,i) = \delta(C_n(t,i-1), C_n(t,i), C_n(t,i+1))$, where $C_n(t,0) = C_n(t,n+1) = *$; it is defined up to the first time at which some cell is F, and statements about C_n refer to these times. The rule *synchronizes* it at time T if all cells are F at time T and none before. A *minimal-time solution* synchronizes the line of length n at time $2n-2$ for every $n \geq 2$; it has k states if $|S| = k$. The *half-line* $C_\infty : \mathbb{N} \times \{1, 2, \dots\} \to S$ is defined in the same way with a border on the left only. Lemmas 2.1–2.3 and 3.1 are elementary; the first two are the light-cone argument behind the bound $2n-2$.

Lemma 2.1. *For every rule, $C_n(t,i) = C_\infty(t,i) = \mathsf{L}$ whenever $t < i-1$ (and $i \leq n$ for C_n).*

Proof. Induction on t; for $t = 0$ this is the initial condition. For $t \geq 1$ the predecessors $(t-1, i')$, $i' \in \{i-1, i, i+1\}$, satisfy $t-1 < i'-1$, so they are L or the right border, and (1) gives L. □

Lemma 2.2 (agreement)**.** *For every rule and $n \geq 2$: $C_n(t,i) = C_\infty(t,i)$ whenever $1 \leq i \leq n$ and $t+i \leq 2n-2$.*

Proof. Induction on t, the case $t = 0$ being clear. If $i = n$, then $t \leq n-2 < i-1$ and both sides are L by Lemma 2.1. If $i < n$, the predecessors $(t-1, j)$, $j \in \{i-1, i, i+1\}$, satisfy $(t-1)+j \leq 2n-2$ and $j \leq n$, so they agree by induction (for $j = 0$ both are the border). □

The border is felt only in the *reflected triangle* $R_n := \{(t,i) : 1 \le i \le n,\ t+i \ge 2n-1,\ t \le 2n-2\}$. If a rule synchronizes the line of length n at a time $T \le 2n-3$, then by Lemma 2.2 the first cell of every line of length $n' \ge n$ fires at time T, while for $n' > T+1$ the cell n' is still quiescent (Lemma 2.1); so a rule that synchronizes every line needs time at least $2n-2$ [3].

Lemma 2.3 (return chain)**.** *If δ synchronizes the lines of lengths n and $n+1$ at times $2n-2$ and $2n$, then $C_n(2n-1-i,i) \neq C_\infty(2n-1-i,i)$ for $1 \le i \le n$, and the* front *$f_i := C_\infty(i-1,i)$ satisfies $f_{n-1} \neq \mathsf{L}$.*

Proof. For $i=1$, $C_n(2n-2,1) = \mathsf{F}$, whereas $C_\infty(2n-2,1) = C_{n+1}(2n-2,1) \neq \mathsf{F}$ by Lemma 2.2 and since $2n-2 < 2n$. If the claim holds for $i < n$, then the left and middle predecessors of $(2n-1-i,i)$ have coordinate sums $2n-3$ and $2n-2$ and agree in C_n and C_∞ (for $i=1$ the left one is the border), so the right ones differ: this is the claim for $i+1$. For $i=n$ it reads $\delta(f_{n-1},\mathsf{L},*) \neq \delta(f_{n-1},\mathsf{L},\mathsf{L})$ (by Lemmas 2.1 and 2.2), which is impossible if $f_{n-1} = \mathsf{L}$ by (1). □

In a minimal-time solution every cell i therefore leaves L exactly at time $i-1$, and since $C_\infty(t,i) = C_n(t,i)$ for n large and the line of length n does not fire before time $2n-2$,

$$C_\infty(t,i) \neq \mathsf{F} \qquad \text{for all } t \ge 0,\ i \ge 1. \tag{2}$$

3. The right end: a speed-1/3 barrier

Fix $n \ge 2$ and use the relative coordinates $\tau := t-(n-1)$, $\kappa := n-i$, in which $R_n = \{(\tau,\kappa) : 0 \le \kappa \le \tau \le n-1\}$, with the right border at $\kappa = -1$ and the left border at $\kappa = n$. The predecessors of a cell of R_n lie in R_n, on a border, or on one of the anti-diagonals $t+i = 2n-3$, $t+i = 2n-2$, where $C_n = C_\infty$. The *input word* of R_n is

$$J_n(\kappa) := \big(C_\infty(n-3+\kappa,\ n-\kappa),\ C_\infty(n-2+\kappa,\ n-\kappa)\big), \qquad 0 \le \kappa \le n-1,$$

with *lower* and *upper input cells* on $t+i = 2n-3$ and $2n-2$ (for $n=2$, $\kappa=0$ the lower cell lies at time -1, is a predecessor of no cell of R_n, and is set to L); see Figure 1.

Lemma 3.1 (determinacy)**.** *There is a partial map Ψ, depending on δ only, such that $C_n(n-1+\tau, n-\kappa) = \Psi(J_n, n; \tau, \kappa)$ for all $n \ge 2$ and $(\tau,\kappa) \in R_n$, and $\Psi(J,m;\tau,\kappa)$ depends only on the input cells and borders in the backward dependency cone of (τ,κ).*

Proof. Define $\Psi(J,m;\tau,\kappa)$ for $0 \le \kappa \le \tau \le m-1$ by induction on τ as $\delta(x_{\kappa+1}, x_\kappa, x_{\kappa-1})$, where $x_{\kappa'}$ is given by the first applicable case: $*$ if $\kappa' \in \{-1, m\}$; the lower or upper component of $J(\kappa')$ if $\tau - 1 = \kappa' - 2$ or $\tau - 1 = \kappa' - 1$; otherwise $\Psi(J,m;\tau-1,\kappa')$ (then $0 \le \kappa' \le \tau-1$). For $m=n$ these are the left, middle and right predecessors of the cell $(n-1+\tau, n-\kappa)$, with $C_n = C_\infty$ on the input cells, so induction on τ gives the first claim; unfolding the definition gives the second. □

For $(\tau,\kappa) \in R_n$ let $\mathrm{cone}_n(\tau,\kappa)$ be the set of input cells reached from (τ,κ) by repeatedly passing to predecessors, stopping at input cells and borders.

Lemma 3.2. *For $n \ge 2$, $\mathrm{cone}_n(n-1,0)$ avoids the left border and consists of the upper input cell at $\kappa' = 0$, both input cells at $1 \le \kappa' \le \lfloor n/2 \rfloor$ and, if n is odd, the lower one at $\kappa' = (n+1)/2$.*

Proof. A step changes τ by -1 and κ by at most one, and input cells satisfy $\tau' \in \{\kappa'-1, \kappa'-2\}$; a path from $(n-1,0)$ reaching an input cell at κ' after r steps has $\kappa' \le r$ and $n-1-r \in \{\kappa'-1, \kappa'-2\}$, so $2\kappa' \le n+1$, and $2\kappa' \le n$ for upper cells. Conversely, the upper cell at $\kappa' \le n/2$ is reached by κ' left moves followed by vertical moves, and the lower cell at $1 \le \kappa' \le (n+1)/2$ by $\kappa'-1$ left moves, vertical moves down to $(\kappa'-1, \kappa'-1)$ and one more left move, through cells of R_n; the lower cell at $\kappa' = 0$ is a predecessor of no cell of R_n. Every cell of R_n on such a path satisfies $\kappa \le \tau \le n-1-\kappa$, hence $\kappa \le (n-1)/2$, so no such cell is adjacent to the left border $\kappa = n$. □

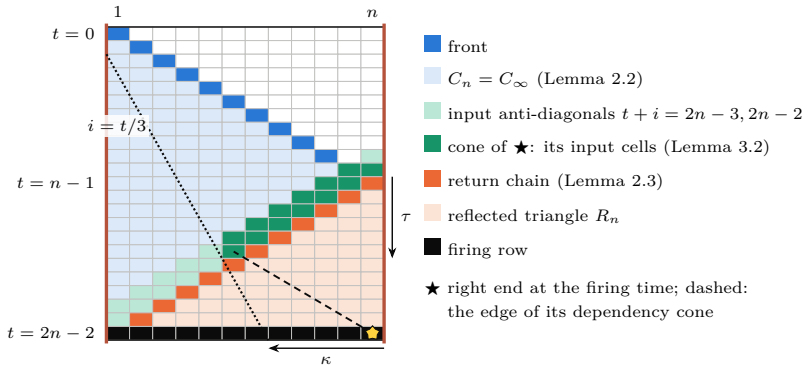


Figure 1: The line of length $n = 12$ (time downwards; colours show regions, not states). R_n is computed from the two input anti-diagonals (Lemma 3.1); the firing of the right end (★) depends only on the dark input cells (Lemma 3.2), which end where the line $i = t/3$ crosses the input anti-diagonals (Theorem 3.4).

Lemma 3.3 (pumping)**.** *Let $2 \le n < n'$ and let δ synchronize the lines of lengths n and n' at times $2n-2$ and $2n'-2$. Then J_n and $J_{n'}$ differ on* $\mathrm{cone}_n(n-1, 0)$.

Proof. Otherwise, by Lemmas 3.1 and 3.2, the relative cell $(n-1, 0)$ of $R_{n'}$ (it lies in $R_{n'}$, and its cone has the same positions, away from the left border $\kappa = n'$) has the value of the relative cell $(n-1, 0)$ of R_n, namely $C_n(2n-2, n) = \mathsf{F}$. Thus $C_{n'}(n'+n-2, n') = \mathsf{F}$ although $n'+n-2 < 2n'-2$. □

We write $\mathrm{PUMP}(n, n')$ for the conclusion of Lemma 3.3; it is a condition on C_∞ alone. In the frame of the front, $\Phi_t(j) := C_\infty(t, t+1-j)$ for $0 \le j \le t$ is the cell at *depth* j behind the front, and the recurrence becomes

$$\Phi_{t+1}(j) = \delta\big(\Phi_t(j),\, \Phi_t(j-1),\, \Phi_t(j-2)\big) \qquad (0 \le j \le t+1), \tag{3}$$

with $\Phi_t(-1) = \Phi_t(-2) = \mathsf{L}$ and $\Phi_t(t+1) = *$: information only travels to larger depths, and the *depth-row* $t \mapsto \Phi_t(j)$ starts at time j. In these coordinates $J_n(\kappa) = (\Phi_{n-3+\kappa}(2\kappa-2), \Phi_{n-2+\kappa}(2\kappa-1))$.

Theorem 3.4 (speed-1/3 barrier)**.** *Let δ be a minimal-time solution. Every depth-row $t \mapsto \Phi_t(j)$ is eventually periodic; let $\theta(j)$ be the least $t_0 \ge j$ such that it is periodic on $[t_0, \infty)$. For every $n \ge 4$ there is a depth $j \le n$ with $\theta(j) > n + \frac{j}{2} - \frac{5}{2}$. Hence $\limsup_{j\to\infty} \theta(j)/j \ge 3/2$: along infinitely many depth-rows the non-periodic part extends, up to an arbitrarily small angle, to the line $i = t/3$.*

Proof. Row 0 satisfies $\Phi_{t+1}(0) = \delta(\Phi_t(0), \mathsf{L}, \mathsf{L})$ and is eventually periodic; by (3), row j is a finite-state system driven by the eventually periodic rows $j-1, j-2$, hence eventually periodic, with some period P_j (cf. [14, §4]). Fix $n \ge 4$ and a common multiple P of $P_0, \dots, P_{n-1}$. The input cells of the cone at $\kappa = 0$ are quiescent in R_n and R_{n+P} (Lemma 2.1); the others have depth $2\kappa-2$ (lower) or $2\kappa-1$ (upper), at most $n-1$ by Lemma 3.2. If each of them lies in the periodic part of its row, i.e. $n-3+\kappa \ge \theta(2\kappa-2)$ for the lower and $n-2+\kappa \ge \theta(2\kappa-1)$ for the upper ones, then $J_{n+P} = J_n$ on the cone, contradicting Lemma 3.3. So some such cell, of depth $j \le n$, has $\theta(j) > n-3+\kappa \ge n + \frac{j}{2} - \frac{5}{2}$. If $\limsup \theta(j)/j < 3/2$, then $\theta(j) \le \alpha j + \beta$ for all j with some $\frac{1}{2} \le \alpha < \frac{3}{2}$ and β, and for n large and all $j \le n$ we get $(\alpha - \frac{1}{2})j \le (\alpha - \frac{1}{2})n < n - \beta - \frac{5}{2}$, i.e. $\theta(j) \le n + \frac{j}{2} - \frac{5}{2}$, a contradiction. Finally, the cell of depth j at time $\theta(j)$ is at position $\theta(j) + 1 - j$, which is at least $\theta(j)/3 + 1$ when $\theta(j) \ge 3j/2$. □

Remark 3.5. The speed-1/3 signal of Waksman's and Balzer's constructions runs along the line $i = t/3$; in Mazoyer's solution the periodic zone behind the front ends at speed 1/2 (observed up to time 2600: $\theta(j) \in \{2j+1, 2j+2\}$ for $1 \le j < 700$, with period 3). Computations suggest that the constant 3/2 is the best that Lemma 3.3 alone can give: on the half-line of the five-state rule δ_{12} of Section 5, observed up to time 1400, $\theta(j) = \lceil 3j/2 \rceil + 2$ for $10 \le j \le 300$, with period 1, and the half-line satisfies $\mathrm{PUMP}(n, n')$ for all $4 \le n < n' \le 650$, but it is excluded by the second barrier (Corollary 5.3).

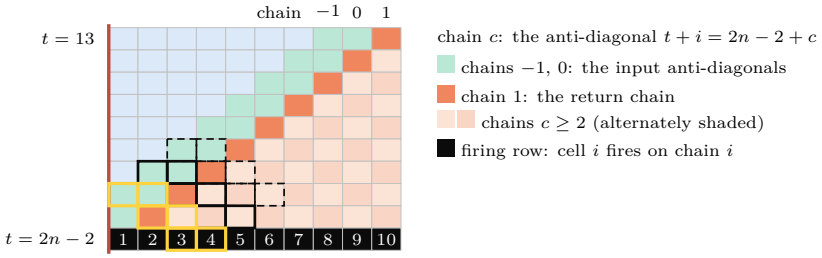


Figure 2: The lower left corner of the line of length $n = 12$ (time downwards; colours show regions, not states). Outlined: three slices for $s = 2$ (dashed $y = 4$, solid $y = 3$, yellow $y = s$), each made of the input cells $u_{-1}(y), u_0(y)$ (upper left) and the band cells $u_1(y), \dots, u_4(y)$; slice y is computed from slice $y+1$ (the map Γ in the proof of Lemma 4.1), and slice s contains the firing cells $2s-1$ and $2s$.

4. The left end: a second barrier

Near the left border the reflected triangle is produced by a band of anti-diagonals behind the reflected wave, which acts like a finite automaton reading the half-line and must fire each cell exactly when it arrives there. Put $v_c(i) := C_n(2n-2+c-i,\, i)$ for $c \geq 0$ and $\max(1, c) \leq i \leq n$, and for $c = -1$ and $1 \leq i \leq n-1$: the cell of the *chain* $t + i = 2n - 2 + c$ at position i. Chains -1 and 0 are the input anti-diagonals and chain 1 is the return chain. The predecessors of $(2n-2+c-i, i)$ lie on the chains $c-2$, $c-1$, c, so

$$v_c(i) = \delta\big(v_{c-2}(i-1),\, v_{c-1}(i),\, v_c(i+1)\big) \qquad (c \geq 1), \tag{4}$$

with $v_{c-2}(0)$ and $v_c(n+1)$ standing for the border. If the line of length n is synchronized at time $2n-2$, then $v_i(i) = \mathsf{F}$ and $v_c(i) \neq \mathsf{F}$ for $c < i$. In the skewed coordinates $u_c(y) := v_c(y + \lfloor c/2 \rfloor)$ this becomes

$$u_c(y) = \delta\big(u_{c-2}(y),\, u_{c-1}(y + \varepsilon_c),\, u_c(y+1)\big), \qquad \varepsilon_c = 1 \text{ for even } c,\ \varepsilon_c = 0 \text{ for odd } c, \tag{5}$$

with $u_0(y) = C_\infty(2n-2-y, y)$ and $u_{-1}(y) = C_\infty(2n-2-y, y-1)$; here $u_c(y)$ is the cell $y + \lfloor c/2 \rfloor$ at time $2n - 2 + \lceil c/2 \rceil - y$, which is F if $\lceil c/2 \rceil = y$ and not F if $\lceil c/2 \rceil < y$. Each slice $(u_1(y), u_2(y), \dots)$ is thus computed from the slice $y+1$ and the input cells $u_{-1}(y), u_0(y)$ (Figure 2). A word w on an integer interval I is *ℓ-periodic* if $w(y+\ell) = w(y)$ whenever $y, y+\ell \in I$.

Lemma 4.1 (left border)**.** *Let δ have k states and synchronize the line of length n at time $2n-2$. Let $s, \ell \geq 1$ and $s \leq Y \leq n - s$, and suppose that the words $y \mapsto C_\infty(2n-2-y, y)$ $(s \leq y \leq Y)$ and $y \mapsto C_\infty(2n-3-y, y)$ $(s \leq y \leq Y-1)$ are ℓ-periodic. Then $Y - s \leq (k-1)^{2s}\ell$.*

Proof. For $s \leq y \leq Y$ let $\mathbf{u}(y) := (u_1(y), \dots, u_{2s}(y))$; these are cells $y + \lfloor c/2 \rfloor \leq n$ with $\lceil c/2 \rceil \leq s \leq y$, so they are defined, and none is F when $y > s$. By (5), $u_c(y)$, $1 \leq c \leq 2s$, is computed from components of index $< c$ of $\mathbf{u}(y)$, of index $\leq c$ of $\mathbf{u}(y+1)$, and from $u_{-1}(y), u_0(y)$; hence $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y))$ for $s+1 \leq y \leq Y-1$, with Γ depending on δ only. Unfolding (5) for even c, $u_{2j}(y) = \delta(u_{2j-2}(y), u_{2j-1}(y+1), u_{2j}(y+1))$, shows that $u_{2s}(y) = \Gamma'(\mathbf{u}(y+1), u_0(y))$ for $s \leq y \leq Y-1$. By hypothesis, u_0 is ℓ-periodic on $[s, Y]$ and u_{-1} on $[s+1, Y]$.

Suppose $Y - s > (k-1)^{2s}\ell$. The pairs $(\mathbf{u}(y), y \bmod \ell)$, $s+1 \leq y \leq Y$, take at most $(k-1)^{2s}\ell$ values, so there are $s+1 \leq y_1 < y_2 \leq Y$ with $\mathbf{u}(y_1) = \mathbf{u}(y_2)$ and $\ell \mid h := y_2 - y_1$. By downward induction, $\mathbf{u}(y) = \mathbf{u}(y+h)$ for $s+1 \leq y \leq y_1$: if it holds for $y+1 \leq y_1$, then $\mathbf{u}(y) = \Gamma(\mathbf{u}(y+1), u_{-1}(y), u_0(y)) = \Gamma(\mathbf{u}(y+1+h), u_{-1}(y+h), u_0(y+h)) = \mathbf{u}(y+h)$, as $y+h < y_2 \leq Y$. Since $s+h \leq Y-1$ and $u_0(s+h) = u_0(s)$,

$$u_{2s}(s+h) = \Gamma'(\mathbf{u}(s+1+h), u_0(s+h)) = \Gamma'(\mathbf{u}(s+1), u_0(s)) = u_{2s}(s) = \mathsf{F},$$

although $\lceil 2s/2 \rceil = s < s+h$, so that $u_{2s}(s+h)$, the cell $2s+h$ at time $2n-2-h$, is not F: a contradiction. □

Theorem 4.2 (left-border barrier)**.** *Let δ be a minimal-time solution. There are no $s \geq 1$, $\alpha > 0$, t_0 and $p, q \geq 1$ such that $C_\infty(t+p, i) = C_\infty(t, i)$ and $C_\infty(t, i+q) = C_\infty(t, i)$ whenever the cells involved lie in the wedge $\Omega := \{(t, i) :\ s \leq i \leq \alpha t,\ t \geq t_0\}$.*

Proof. Suppose otherwise and put $\ell := pq$. If (t,i) and $(t-\ell,i+\ell)$ lie in Ω, so do the cells $(t-\ell,i+jq)$, $0 \le j \le p$, and $(t-\ell+jp,i)$, $0 \le j \le q$, hence $C_\infty(t-\ell,i+\ell) = C_\infty(t-\ell,i) = C_\infty(t,i)$. Let $Y_n := \lfloor \alpha(2n-3)/(1+\alpha) \rfloor$. For $s \le y \le Y_n$ the cells $(2n-2-y,y)$ and $(2n-3-y,y)$ satisfy $y \le \alpha(2n-3-y)$, so they lie in Ω once $2n-3-Y_n \ge t_0$, i.e. for n large; since a step of ℓ positions along an anti-diagonal is the step from (t,i) to $(t-\ell,i+\ell)$, the words of Lemma 4.1 are ℓ-periodic on $[s,Y_n]$ and $[s,Y_n-1]$. As $Y_n \ge \frac{2\alpha}{1+\alpha} n - 4$, for n large $Y := \min(Y_n, n-s)$ satisfies $Y - s > (k-1)^{2s}\ell$, contradicting Lemma 4.1. □

The half-line of a rule is *eventually regular* if there are t_1, $P \ge 1$ and, for each residue $r \in \{0,\dots,P-1\}$, an integer m, words $w_0,\dots,w_m$, nonempty words $z_1,\dots,z_m$ over W and integers $a_j, D_j \ge 0$ (all depending on r) such that for all $\nu \ge 0$ the configuration $C_\infty(t,1)\cdots C_\infty(t,t+1)$ at time $t = t_1 + r + \nu P$ is $w_0 z_1^{a_1+\nu D_1} w_1 \cdots z_m^{a_m+\nu D_m} w_m$: finitely many periodic regions separated by signals of rational speeds.

Corollary 4.3. *The half-line of a minimal-time solution is not eventually regular.*

Proof. Suppose it is. For each residue r some D_j is positive, because the configuration grows by P cells per period; let $j(r)$ be the least such index. The blocks and junctions left of $z_{j(r)}$ have constant lengths, so the block $z_{j(r)}^{a_{j(r)}+\nu D_{j(r)}}$ starts at a fixed cell s_r and its length grows at least like βt, for some $\beta > 0$ and t large. Inside it the state of a cell depends only on its distance to s_r modulo $|z_{j(r)}|$, and is the same at the times t and $t+P$. With $s := \max_r s_r$, $q := \prod_r |z_{j(r)}|$, $\alpha := \beta/2$ and t_0 large, every cell of $\{(t,i) : s \le i \le \alpha t,\ t \ge t_0\}$ lies in the block of index $j(r)$ of its residue class, so C_∞ is periodic with periods P in time and q in space on this wedge, contradicting Theorem 4.2. □

Remark 4.4. The proof of Lemma 4.1 uses only (4) for $c \ge 1$, the firing condition and the periodicity of the chains $c = -1, 0$. Applied to the mirror image of the line, it shows: if δ synchronizes the line of length n at time $2n-2$, $s \le Y \le n-s$ and $Y - s > (k-1)^{2s}\ell$, then the words $y \mapsto C_n(2n-2-y, n+1-y)$ $(s \le y \le Y)$ and $y \mapsto C_n(2n-3-y, n+1-y)$ $(s \le y \le Y-1)$ are not both ℓ-periodic; for $2y < n$ these cells lie in R_n. In the classical constructions the structure required by Theorem 4.2 is the family of signals of speeds $1/(2^j-1)$ accumulating at the border [3, 6], or the recursive division points [8].

5. Applications to four and five states

The CNF formula $\mathrm{MT}_k(\mathcal{N}; M, N')$ has one-hot variables for the entries of a k-state rule table, for the cells of C_∞ with $t + i \le M$, and for the cells of R_n, $n \in \mathcal{N}$. Its clauses say that these cells follow the rule, that no half-line cell fires (2), that the line of length $n \in \mathcal{N}$ fires at time $2n-2$ and not before, and that the return chains of Lemma 2.3 differ from C_∞; they add $\mathrm{PUMP}(n, n')$ for $4 \le n < n' \le N'$ ($\mathrm{MT}_k(\mathcal{N}; M)$ has none). All families follow from the definition of minimality and the lemmas above, so the formula is satisfiable if a k-state minimal-time solution exists. (An optional symmetry breaking, by first occurrence of the auxiliary states in time-major order, is used by none of the formulas below.) As a test, Mazoyer's solution, extracted mechanically from Duprat's Coq development [21] whose correctness proof is described in [22], satisfies $\mathrm{MT}_6(\{3,\dots,12\}; 78, 40)$ (the extracted table synchronizes the lengths $n \ge 3$), and every model found by a solver was re-simulated by an independent checker. All claims of this section are computer-assisted: the unsatisfiability of each generated formula is certified by an LRAT proof [23], obtained from CaDiCaL [24] or from Kissat [25] via `drat-trim` and checked by `lrat-check` [26], whereas the formula generator, the enumeration and the simulations are trusted programs, tested as described.

Theorem 5.1. *No four-state rule satisfying* (1) *synchronizes all lines of lengths* $2,\dots,9$ *in minimal time; in particular no four-state minimal-time solution exists.*

Proof. Such a rule yields a model of $\mathrm{MT}_4(\{2,\dots,9\}; 16)$: a half-line cell with $t + i \le 16$ is quiescent if $i \ge 10$ (Lemma 2.1) and equals a cell of the line of length 9 at a time $t < 16$ if $i \le 9$ (Lemma 2.2), so it does not fire, and the return chains hold by Lemma 2.3 for $2 \le n \le 8$. CaDiCaL refutes this formula (753 variables, 17,605 clauses, SHA-256 `d18d65480e77505193b8f60362badad5e71f8f08505f29befbcd3b2b18cdaabf`) in 51 s with an LRAT proof of 177 MB, which `lrat-check` verifies in 5.7 s. □

x	y	$*$	L	G	A	B	x	y	$*$	L	G	A	B
$*$	L	–	F	F	L	G	$*$	A	–	G	G	L	G
L	L	L	L	G	G	B	L	A	A	B	G	B	A
G	L	G	A	B	A	B	G	A	B	A	G	L	A
A	L	G	B	G	G	G	A	A	B	L	A	G	A
B	L	A	B	A	A	L	B	A	L	A	A	G	A
$*$	G	–	L	F	L	L	$*$	B	–	G	G	G	G
L	G	F	A	F	B	A	L	B	G	B	G	A	L
G	G	B	B	F	G	F	G	B	F	B	A	B	B
A	G	B	A	G	G	A	A	B	G	B	B	G	B
B	G	B	A	G	A	L	B	B	A	L	A	A	L

Table 1: The rule δ_{14} of Proposition 5.2: the entry in row (x, y) and column z is $\delta(x, y, z)$; "–" marks the unused neighbourhoods $(*, y, *)$. Machine-readable copy: `results/delta14.txt` in the repository.

The bound 9 is sharp: an exhaustive enumeration finds 27 partial rules (rules restricted to the transitions they use) for the lengths $2, \dots, 8$. Yunès [9], who also imposes $\delta(*, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, found 27 as well; none of ours uses $(*, \mathsf{L}, \mathsf{L})$. Sanders' example rule for the lengths up to 8 [10] is one of them.

The five-state frontier. A five-state table has 94 free entries with five values, against 43 entries with four values for four states.

Proposition 5.2. *The five-state rule δ_{14} of Table 1 synchronizes the line of length n at time $2n - 2$ for every $2 \le n \le 14$; on the line of length 15 the cells 13, 14, 15 fire at time 22. Its half-line uses 16 neighbourhoods besides $(\mathsf{L}, \mathsf{L}, \mathsf{L})$ on the cells with $t + i \le 10^4$, none of them firing, and satisfies $\mathrm{PUMP}(n, n')$ for $4 \le n < n' \le 800$; for $n \le 5000$, $s \le 3$ and $\ell \le 24$, Lemma 4.1 does not apply to its input anti-diagonals.*

Proof. Direct simulation, by programs independent of the encoder. □

For $3 \le t < 1500$ the half-line of δ_{14} takes the values L, G at cell 1, L, A at cell 2 and L, B at the cells $i \ge 3$; the rule restricted to $\{\mathsf{L}, \mathsf{B}\}^3$ is the elementary cellular automaton rule 22 [27], so the half-line is rule 22 driven at its left end by the cells 1 and 2, and it is irregular. Four further rules, on three other half-lines (up to renaming of states), also synchronize all lengths up to 14 and fail at 15. Hence $\mathrm{MT}_5(\{2, \dots, 14\}; 78, 40)$ is satisfiable, neither barrier excludes δ_{14} in the ranges of Proposition 5.2, and refutations along these lines need lines of length at least 15.

The left-border lemma does refute a more regular architecture. The rules δ_{12} and δ_{13} (files `results/uniformA_2-12.txt` and `results/delta13.txt`) synchronize every line of length at most 12, resp. 13. Their half-lines coincide for $t + i \le 78$, where they use the 22 transitions

```
*GL>A GLL>G *AG>B AGL>L *BL>A BLG>L LGL>B *AL>G ALB>A LBG>L BGL>G
*GA>A GAL>L ALG>B LGG>G GGL>G BGG>L GGG>G ALL>A LLG>L ALA>A LAL>L
```

(where `xyz>d` means $\delta(x, y, z) = d$) besides $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$, and consist of a front and a wake of G's and, behind a boundary of speed 1/3, a region $\mathsf{A}\,\mathsf{L}\,\mathsf{A}\,\mathsf{L}\cdots$ periodic in time and space (Remark 3.5).

Corollary 5.3. *No five-state rule with the 22 transitions above synchronizes the line of length 518 in minimal time.*

Proof. With these transitions and $\delta(\mathsf{L}, \mathsf{L}, \mathsf{L}) = \mathsf{L}$ only, no other neighbourhood occurs on the cells with $t + i \le 1034$, so every such rule has the same diagram C_∞ there. On $t + i = 1034$ the cells $2, \dots, 259$ are in state A, and on $t + i = 1033$ the cells $2, \dots, 258$ are in state L. Lemma 4.1 with $k = 5$, $n = 518$, $s = 2$, $\ell = 1$, $Y = 259$ would give $257 \le 4^4 = 256$. □

Half-lines of low complexity. The *germ* below M of a rule is its restriction to the neighbourhoods of the cells (t, i), $t \geq 1$, $t + i \leq M$, of its half-line, and its *complexity* $c_M(\delta)$ is the number of these neighbourhoods other than $(\mathsf{L}, \mathsf{L}, \mathsf{L})$: $c_{78} = 57$ for Mazoyer's rule, 22 for δ_{12} and 16 for δ_{14}.

Proposition 5.4. *Every five-state minimal-time solution δ satisfies $c_{78}(\delta) \geq 15$.*

Proof (computer-assisted; the SAT part is LRAT-certified). Suppose $c_{78}(\delta) \leq 14$. After exchanging A and B if necessary, A occurs before B (if at all) when the cells are ordered by anti-diagonal and, within an anti-diagonal, by time. By (2) and Lemmas 2.3 and 3.3, the germ of δ below 78 produces no firing cell, never produces L at the front, and satisfies $\mathrm{PUMP}(n, n')$ for $4 \leq n < n' \leq 40$. A depth-first search that chooses the value of a neighbourhood when it first occurs enumerates all such germs ($1.0 \cdot 10^8$ nodes): there are 244, and an independent re-implementation finds the same 244. For 230 of them, $\mathrm{MT}_5(\{2, \ldots, 10\}; 78, 39)$ (without symmetry breaking) together with the germ is unsatisfiable: one formula with a selector variable per germ, clauses that make each selector fix the transitions of its germ, and a clause requiring some selector, is refuted by CaDiCaL in 4 s with an LRAT proof of 11 MB. Each of the other 14 germs determines C_∞ up to the anti-diagonal 3300, and on these diagrams Lemma 4.1 applies with $s = 1$: there are $n \leq 163$ and $\ell \leq 5$ such that, for the line of length n, the upper input anti-diagonal is ℓ-periodic on $[1, Y]$ and the lower one on $[1, Y - 1]$ with $Y - 1 > 16\ell$ (e.g. $n = 28$, $\ell = 1$, $Y = 19$), as an independent simulation confirms. Hence no rule containing one of the 244 germs, in particular δ, is a minimal-time solution. □

Balzer's conditions. Balzer [6, p. 37] reported that no five-state minimal-time solution satisfies the four conditions (B1) $\delta(x, \mathsf{G}, y) = \mathsf{G}$ unless $x, y \in \{\mathsf{G}, *\}$; (B2) $\delta(I(z), I(y), I(x)) = I(\delta(x, y, z))$ for an involution I of the states with $I(*) = *$ (an *image solution*); (B3) $\delta(\mathsf{G}, \mathsf{G}, \mathsf{G}) = \delta(*, \mathsf{G}, \mathsf{G}) = \delta(\mathsf{G}, \mathsf{G}, *) = \mathsf{F}$ and no other neighbourhood gives F; (B4) $\delta(\mathsf{G}, v, \mathsf{G}) = \mathsf{G}$ for all $v \in W \setminus \{\mathsf{G}\}$. His eight-state solution satisfies them, and Mazoyer's only (B3) [28]. In the *weak reading* [9], (B1) is required only for working states x, y and (B3) without its last clause. We take $I(\mathsf{F}) = \mathsf{F}$. Then $I(\mathsf{G}) = \mathsf{G}$ in both readings: otherwise $g := I(\mathsf{G}) \in W \setminus \{\mathsf{G}\}$, (B4) gives $\delta(\mathsf{G}, g, \mathsf{G}) = \mathsf{G}$, hence $\delta(g, \mathsf{G}, g) = g$ by (B2), whereas (B1) gives $\delta(g, \mathsf{G}, g) = \mathsf{G}$. So I is one of the four involutions of $\{\mathsf{L}, \mathsf{A}, \mathsf{B}\}$, and (B1)–(B4) become unit and binary clauses.

Proposition 5.5. *No five-state rule satisfying (B1)–(B4) in the weak reading, with $I(\mathsf{F}) = \mathsf{F}$, synchronizes all lines of lengths $2, \ldots, 12$ in minimal time; a fortiori none satisfying Balzer's (strong) form does. The lengths up to 11 do not suffice to prove this.*

Proof (computer-assisted; the SAT part is LRAT-certified). For each of the four involutions, the clauses of (B1)–(B4) in the weak reading are added to $\mathrm{MT}_5(\{2, \ldots, N\}; 2N - 2)$, without symmetry breaking. For $I = (\mathsf{A}\,\mathsf{B})$ and $N = 10$, and for $I = (\mathsf{L}\,\mathsf{A})$ and $I = (\mathsf{L}\,\mathsf{B})$ and $N = 9$, CaDiCaL refutes the formula with LRAT proofs checked in 7.1, 3.5 and 3.3 s. For I the identity and $N = 12$, Kissat refutes it in 15 minutes; `drat-trim` turns its DRAT proof into an LRAT proof of 3.1 GB, which `lrat-check` verifies in 66 s. For the last statement, the rule in `results/balzer_sym11.txt` satisfies (B1)–(B4) in the strong, hence also in the weak, reading with I the identity, and it synchronizes every line of length at most 11 (direct simulation). □

With six states, for every type of involution there are rules satisfying (B1)–(B4) in the strong reading that synchronize all lines of lengths up to 13, and for two types up to 14, so a refutation of this kind would need longer lines; whether Balzer's conditions exclude six states remains open.

6. Concluding remarks

A minimal-time solution needs non-periodic structure reaching the line $i = t/3$ and in every wedge at the left border; the known constructions provide it with hierarchies of slow signals or division points, while the five-state half-lines that survive our tests are irregular, and Lemma 4.1 has no grip on them in the ranges we checked. Natural questions are: (1) Does a five-state minimal-time solution exist? A refutation along the lines of Theorem 5.1 needs lines of length at least 15. (2) Is there a necessary condition like Lemma 3.3 on the reflected triangles themselves, beyond Remark 4.4? (3) Can the bound $(k - 1)^{2s}\ell$ of Lemma 4.1 be made polynomial in s, or extended to anti-diagonals that are periodic up to a bounded number of defects? (4) Do Balzer's conditions exclude six states?

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